

Literature Status Note for arXiv:2510.12540

Run fa_banach_001

2026-07-03

Status. `literature_already_answered`, scoped to the source paper's conjectured strict upper bound for the two-variable universal commuting dilation constant. This note is not a new proof from this run.

Source Question

The source paper is M. Gerhold, M. Scherer, and O. M. Shalit, *Empirical bounds for commuting dilations of free unitaries and the universal commuting dilation constant*, arXiv:2510.12540.

On page 1, the source identifies the open problem of determining the universal commuting dilation constant C_2 , the least constant c such that every pair of contractions dilates to commuting normal operators of norm at most c . The source gives numerical evidence for

$$C_2 \leq 2\sqrt{\frac{2}{3}} < 2.$$

On page 3, after deriving this inequality semi-rigorously from an empirical random-unitary limit, the source explicitly states that this inequality is conjectured to hold.

Supporting Answer

The supporting paper is I. Thompson, arXiv:2606.12506. Its title is *On the universal commuting dilation constant*.

The abstract on page 1 states that the paper gives a positive answer to the question whether $C_2 < 2$. Page 2, Theorem A, proves that for every pair of contractions T_1, T_2 there is a dilation to commuting normal contractions with scale

$$\frac{2}{\sqrt{\varphi}},$$

where φ is the golden ratio. Equivalently,

$$C_2 \leq \frac{2}{\sqrt{\varphi}}.$$

The same page explicitly discusses the 2025 empirical approach and says that the new upper bound is stronger than the suggested bound $C_2 \leq 2\sqrt{2/3}$.

Since $\varphi = (1 + \sqrt{5})/2$, one has

$$\frac{2}{\sqrt{\varphi}} \approx 1.5724 < 2\sqrt{\frac{2}{3}} \approx 1.6330.$$

Thus the supporting theorem proves the source's conjectured inequality and in particular proves $C_2 < 2$.

Why This Is Literature, Not A New Result

The supporting paper explicitly cites and discusses the 2025 empirical approach and states that its main upper bound is stronger than the suggested $2\sqrt{2/3}$ bound. The identification is therefore a direct later-literature answer, not an agent-derived implication.

Scope Limitations

This packet does not claim the exact value of C_2 . It also does not prove the source's empirical random-matrix limit $\lim_{N \rightarrow \infty} c(U^{(N)}, u_0) = \sqrt{2}$. Finally, it does not determine the exact value of C_d for every d ; the supporting paper improves general bounds but does not settle all constants.

Search And Verification Notes

The run indexes were searched for the source arXiv id, title terms, and core keywords including commuting dilations, free unitaries, universal commuting dilation constant, and C_2 . No prior packet for arXiv:2510.12540 was found. Web search found arXiv:2606.12506, posted June 2026, with the stronger Theorem A bound.

Local Files

- Source PDF: `source_paper.pdf`.
- Supporting PDF: `supporting_paper_2606.12506.pdf`.
- Ledger record: `2510.12540_c2_bound_answered_by_2606.12506.json`.

References

- [1] M. Gerhold, M. Scherer, and O. M. Shalit, *Empirical bounds for commuting dilations of free unitaries and the universal commuting dilation constant*, arXiv:2510.12540.
- [2] I. Thompson, *On the universal commuting dilation constant*, arXiv:2606.12506.